

Is there a Hamiltonian inside a decision model?

Probing the energy landscape of Jev for quantum chemistry

Zhaohe Dong
Independent researcher
dongzhaohe321418@gmail.com

Abstract

Jev is a proprietary “System One” model that answers many typed questions about one input in a single pass and returns a calibrated probability for every allowed answer. Any such output defines local energies $E = -\log p$; we ask whether they form one Hamiltonian-like landscape and whether that landscape is useful for quantum chemistry. Using black-box access to `jev-1.13.0`, PySCF ground truth and pre-registered analyses (with disclosed deviations), we find that stated facts shift Jev’s answers approximately additively in log-odds, so single-fact conditionals predict held-out two-fact conditionals in all 48 cases, yet these effects are not symmetric. Under a coherent null matched to Jev’s own noise, 41% of pairwise coupling asymmetries are significant after false-discovery-rate control, the incoherence persists when the exact joint distribution is written into the input, and a symmetric Ising energy predicts no better than an asymmetric dependency network. Given a Hamiltonian as text, Jev follows the signs of the printed coefficients: writing the same Ising energies in the opposite sign convention reverses its preferences, and for the Jordan–Wigner Hamiltonian of H_2 it keeps choosing the textbook Hartree–Fock state after the qubits are relabelled. On quantum-chemistry diagnostics Jev reaches AUROC 0.76–0.89, but most of this comes from a stretch factor stated in the prompt; removing that note drops it to 0.57–0.71. It adds a small amount of information to a two-descriptor model (significant for two of three labels, and for one on a larger set of 31 less common molecules), ranks CCSD(T) failures about as well as the T_1 diagnostic, and ranks determinants for selected CI with one template that loses to iterative CIPSI. Two open-weight language models are less coherent still and ignore the Hamiltonian. Jev’s probabilities are a fast triage signal whose local energies approximately cohere; they do not form a physical energy landscape.

1 Introduction

A classifier that outputs a probability distribution $p(y \mid x)$ can always be rewritten as a Gibbs distribution $p(y \mid x) \propto \exp[-E(x, y)]$ with $E = -\log p$ up to a constant [1, 2]. For one softmax head the rewriting says nothing new. It says more when a model answers *several* questions about the same input, because the per-question energies may or may not assemble into one landscape over joint answers, as the conditionals of a Boltzmann machine do [3]. Only a single landscape can be sampled, minimised or added to a physical Hamiltonian, the uses that Ising and QUBO encodings make routine [4, 5].

Jev [6, 7] makes this question concrete. It accepts text or structured state and a set of typed questions (a yes/no *Noul*, a *Choice* over at most 255 options, or a *Score* over 2–10 ordered levels) and returns a probability for every allowed answer, trained, according to its developer, for calibration against outcomes rather than for human preference. Questions are evaluated in parallel, latency is well under a second, and outputs cannot fall outside the schema [8]. A quantum-chemistry workflow

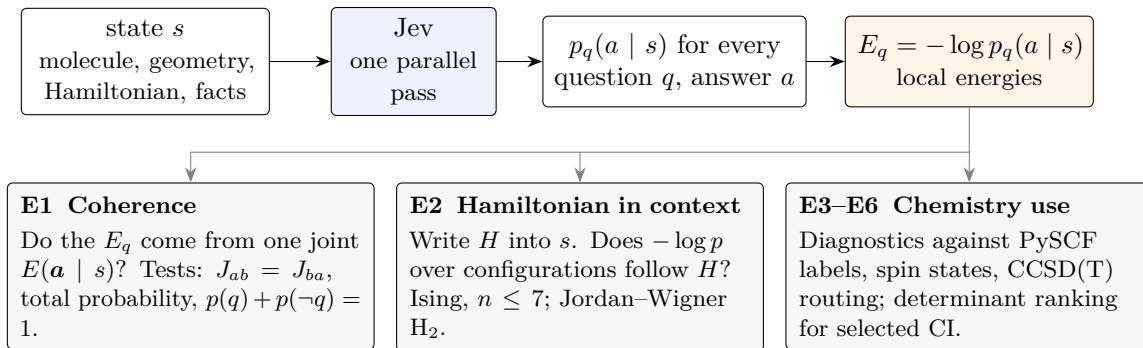


Figure 1: Study design. Jev maps a state and a set of typed questions to answer distributions in one pass. Reading each distribution as a local energy, we test whether the local energies are consistent with one landscape (E1), whether they reproduce a Hamiltonian supplied in the input (E2), and whether they are useful for quantum-chemistry decisions (E3–E6). E7 repeats E1 and E2 on open-weight language models.

makes many small typed decisions of this kind before any expensive calculation: is the system multireference, which spin state is lowest, which determinants matter. The developer also warns that structural invariants “aren’t guaranteed” and that the model “is not a calculator” [9].

We ask three questions and answer each with measurements. (i) Do Jev’s conditionals admit a single joint landscape? We test coupling symmetry and total probability, the classical compatibility conditions for conditionals [10, 11], against a coherent null matched to Jev’s own noise, with a positive control in which the exact joint is written in the input. (ii) Does Jev represent a Hamiltonian written into its input? We give it random Ising models and the Jordan–Wigner qubit Hamiltonian of H_2 [12, 13], and change only how they are written. (iii) Is it useful for quantum-chemistry decisions? We score its predictions of multireference diagnostics against PySCF labels with proper scoring rules [14], ablate its inputs, and use it to rank determinants for selected configuration interaction [15, 16]. After review we added a larger and less familiar set of molecules, spin-state and CCSD(T)-routing tasks, and the same probes on two open-weight language models. Figure 1 summarises the design.

We find that stated facts shift Jev’s local energies roughly additively but asymmetrically, so they do not form one landscape, that its reading of a Hamiltonian follows printed signs and textbook recall rather than energies, and that its chemistry skill mostly reflects a stretch factor stated in the prompt. The coherence tests need only black-box access and apply to any model that answers many questions about one input. Code, cached API responses, the review reports described in Section 3 and figures are released.

2 Background and related work

Classifiers as energy models. Energy-based models assign a scalar energy to each configuration and convert it to probability through a Gibbs distribution [1]. Hopfield networks and Boltzmann machines give such energies pairwise, symmetric couplings [3, 17], and attention can be read as the update of a continuous Hopfield energy [18]. Grathwohl et al. [2] showed that the logits of any softmax classifier define a joint energy over inputs and labels, and Winer and Hanin [19] studied a random network as a Hamiltonian over its inputs. These works treat one output head; Jev exposes many typed heads over one shared state, which is the setting where a joint landscape can fail to exist.

When do conditionals have a joint? Local conditional specifications define a joint distribution only under compatibility conditions [10, 11, 20]. Separately learned conditionals, as in dependency networks, are generally incompatible [21]. For binary variables compatibility has a simple necessary consequence: the log odds ratio between two variables is symmetric, so the coupling inferred from b ’s effect on a must equal that inferred from a ’s effect on b . Recent work finds logical incoherence in the probabilities of language models and forecasters [22–24] and proposes repairs [25, 26]; these studies test identities on generative models, not the joint structure of a parallel, native-distribution model.

Calibration. Deep networks are often miscalibrated [27]; we report expected calibration error [28] and the Brier score [29], a strictly proper rule [14]. Large language models are reasonably calibrated on some formats [30, 31]. Calibration of each marginal does not imply joint coherence, which is why we test both.

Hamiltonians and quantum chemistry. Molecular Hamiltonians are mapped to qubits by the Jordan–Wigner transformation [12] and solved variationally [13, 32], on annealers [33, 34], or with neural wavefunctions [35–37]; see [38, 39] for reviews. Learning a Hamiltonian from data assumes the data come from a Gibbs state [40–42]. Before a calculation, practitioners decide whether a system is multireference, using diagnostics such as T_1 [43, 44], choose active spaces [45, 46], and select determinants [15, 16, 47]. Machine-learned models for these decisions are task-specific and trained on computed descriptors [48–50]. General language models are used in chemistry as tool-using agents [51, 52], where they are knowledgeable but overconfident [53] and fragile on numerical reasoning [54].

3 Setup

Access and energies. We queried `jev-1.13.0` through the public API (which exposes only the alias `jev-latest`; every response records the concrete version) and cached every request and response. Probabilities are returned to two decimals and exact zeros occur, so we clip to $[0.005, 0.995]$ before taking logarithms, define $E_q(a \mid s) = -\log p_q(a \mid s)$, and write $\ell = \text{logit } p$. Identical requests return different probabilities: the repeat standard deviation of a Noul, estimated without bias from triplicate calls, is 0.020. Every query in the main experiments was therefore repeated two or three times and averaged. Median latency over all 10,145 calls was 0.81 s. Ground truth was computed with PySCF [55] and qubit Hamiltonians with OpenFermion [56].

Chemistry ground truth. Twelve molecules (H_2 , LiH , HF , N_2 , F_2 , CO , BH , BeO , C_2 , H_2O , NH_3 , CH_4) were placed at 1.0, 1.5, 2.0 and 2.5 times their equilibrium bond lengths (48 cases). With PySCF and the cc-pVDZ basis we followed RHF to an internally stable minimum, ran a complete-active-space CI in the full valence space with a singlet constraint, frozen-core CCSD, and a multi-start UHF search with stability following. Three labels follow: *multireference* if the weight of the RHF determinant in the CASCI state is below 0.9 (positive rate 0.65), T_1 if the CCSD T_1 diagnostic [43] exceeds 0.02 (0.64; one case without converged CCSD is excluded), and *RHF unstable* if the lowest UHF solution lies more than 1 mE_h below RHF (0.77). Each prompt gives the molecule name, Cartesian geometry, basis, and a note stating the stretch factor.

Pre-registration and deviations. The analyses of E1–E4 were committed to the repository one minute before the first E1 query, after a nine-call pilot. Several things then changed, and we list

them because they affect how the results should be read. Two of the five planned E1 statements (open-shell ground state; excited state within 1 eV) were replaced before collection by the T_1 and RHF-instability statements, so that E1 and E3 share labelled statements. A pre-release review of a first version of this paper, consisting of three mutually blind referee reports simulated by separate AI agents and a code audit by a second AI system (OpenAI Codex), found an error in our first bootstrap, defects in our first chemistry labels (RHF saddle points, active spaces that split degenerate shells, triplet roots, unconverged CCSD), a leak in case-level cross-validation, and an unfair tie-breaking rule in E4. All of these are corrected here, and the first versions remain in the repository. The plan named about 60 E3 cases; we used 48. Its primary E1 criterion, the fraction of pairs whose asymmetry exceeds three bootstrap standard errors, is reported below as a descriptive number, because a pointwise bootstrap over three repeats is not a calibrated test; the E4 area metric and its Wilcoxon tests were not in the plan. The positive control and the inverse temperature were planned and are now reported; everything marked *added after review* was not.

4 Results

4.1 E1: the conditionals are approximately additive but not coherent

For five statements about each case (multireference; RHF qualitatively correct; $T_1 > 0.02$; RHF unstable; CCSD(T) within 1 kcal/mol of full CI) we measured each marginal $p(a)$ and each conditional $p(a | b = v)$, stating b as an “established fact” in the input. If the answers came from one joint distribution, the pairwise coupling

$$J_{ab} = \ell(p(a | b=1)) - \ell(p(a | b=0)) \quad (1)$$

would equal J_{ba} , because both equal the log odds ratio of the pair, and the law of total probability would hold.

Couplings are large (median $|J| = 1.10$ nats) and have the expected signs, but they are not symmetric (Fig. 2a). For each case and pair we built the joint implied by Jev’s own $p(b)$ and $p(a | b)$, simulated the measurement of both directions with Jev’s noise, rounding and repeats, and obtained a p-value for the observed asymmetry. Under this matched coherent null the p-values would be uniform; instead they pile up near zero (Fig. 2b), and 41% of the 480 case–pair asymmetries remain significant after Benjamini–Hochberg control [57] (95% CI 35–48%, bootstrap over molecules). The pre-registered criterion, recomputed after an audit found that our first bootstrap shared repeats across separate calls, gives 53% of asymmetries above three bootstrap standard errors. The pair whose relation comes closest to a logical constraint, “multireference” and “RHF adequate”, also fails: its couplings agree on average (-2.43 against -2.35), but case by case its median asymmetry is 0.55 nats and 60% of its cases are significant. The law of total probability fails by a median of 0.041, and statements asked together with their explicit negations sum to between 0.89 and 1.25.

Two controls were added after review. In the first, Jev did not fully adopt a stated fact: when a statement is declared true in the input, Jev gives that same statement a probability of only 0.80–0.91, and 0.10–0.17 when it is declared false. Our conditionals are therefore soft conditionals, and part of the asymmetry may come from uneven uptake of facts. In the second, a positive control, we stated an explicit population of 1000 people with three binary attributes, so that every conditional is computable and a coherent reader has zero asymmetry. Jev’s asymmetry there was larger than for the chemistry statements (median 0.42 nats; 67% of the 72 table–pair asymmetries significant under the matched null with Benjamini–Hochberg control), and its conditionals were off by 0.10 on average: incoherence persists when the correct joint is written in the input. The result is

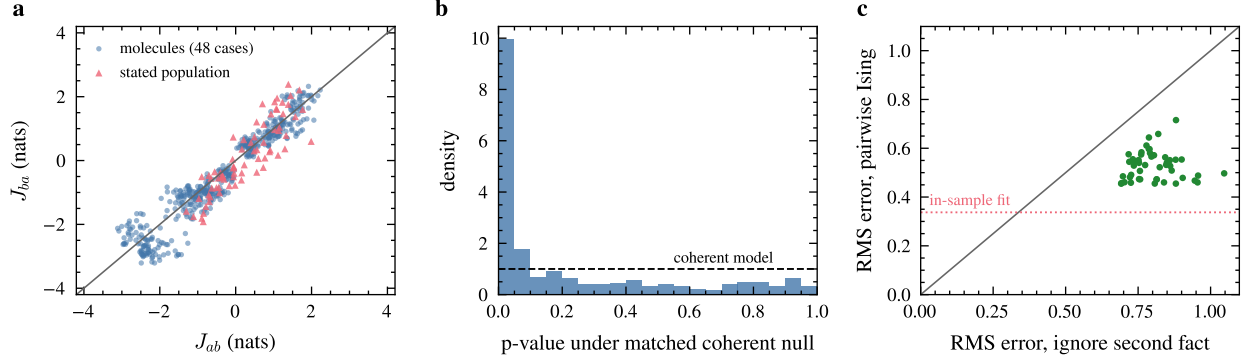


Figure 2: Coherence of Jev’s conditionals (E1). **a**, Coupling inferred in each direction for the 48 molecular cases (circles) and for the stated-population positive control (triangles); a joint distribution requires points on the diagonal. **b**, Per-pair p-values under a coherent null matched to Jev’s own probabilities, noise and rounding; a coherent model gives the dashed uniform density. **c**, Error (logit units) of predicting held-out two-fact conditionals with a pairwise Ising energy fitted per case, against ignoring the second fact; each point is a case, and the dotted line is the in-sample fit error. An asymmetric additive model (not shown) predicts as well as the Ising energy.

not sensitive to wording: an independently phrased set of the five statements gave couplings that correlate at $r = 0.96$ with the originals and a similar asymmetry (median 0.30 nats).

We also asked whether a pairwise Hamiltonian describes the answers (added after review). For each case we fitted one pairwise Ising energy over the five statements to all 165 measured marginals and conditionals, including conditionals on two facts at once. The fit leaves a residual of 0.34 logit units, about five times the measurement noise, so no single pairwise Hamiltonian reproduces the answers. Fitted on marginals and single-fact conditionals only, the Ising energy predicts the held-out two-fact conditionals with an error of 0.53 logit units, against 0.79 for ignoring the second fact, in all 48 cases (Fig. 2c). An asymmetric model that simply adds the two measured single-fact shifts in log-odds, with no shared energy, predicts them equally well (0.52; better than Ising in 22 of 48 cases, Wilcoxon $p = 0.58$). What the held-out test shows is that facts combine roughly additively, not that one landscape exists. Path independence of the three-variable chain rule, a direct test of a joint, is violated on average (median spread across orderings 0.98 nats against 0.39 under the coherent null), but noise accumulates along the chain and only 7% of triples exceed the null’s 95th percentile, so this test is underpowered. Jev’s local energies therefore behave like those of a dependency network [21]: roughly additive, close to symmetric on average, and measurably incoherent case by case.

4.2 E2: a Hamiltonian in the input is read by its signs and by recall

We gave Jev random classical Ising Hamiltonians $H(\mathbf{s}) = \sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j$ with $h_i, J_{ij} \sim U[-1, 1]$ on $n = 3-7$ spins (ten instances per n) and asked, as a Choice over all 2^n configurations, which one minimises H . Jev preferred high-energy configurations: averaged over instances, the Spearman correlation between $p(\mathbf{s})$ and $-H(\mathbf{s})$ ranged from -0.20 to -0.41 (Fig. 3a), and top-1 accuracy was at or below chance for every n (Fig. 3b). Decomposing the energy shows that p correlates with $+\sum_i h_i s_i$ ($\rho = 0.31-0.52$) and little with the coupling term: Jev aligns each spin with the sign of its printed field. The maximum-likelihood inverse temperature of a Gibbs fit, $p \propto e^{-\beta H}$, was negative and small ($\beta = -0.30$ to -0.05). Two alternative wordings of the question gave the same negative correlations.

We then wrote exactly the same energies in the textbook convention $H = -\sum h'_i s_i - \sum J'_{ij} s_i s_j$

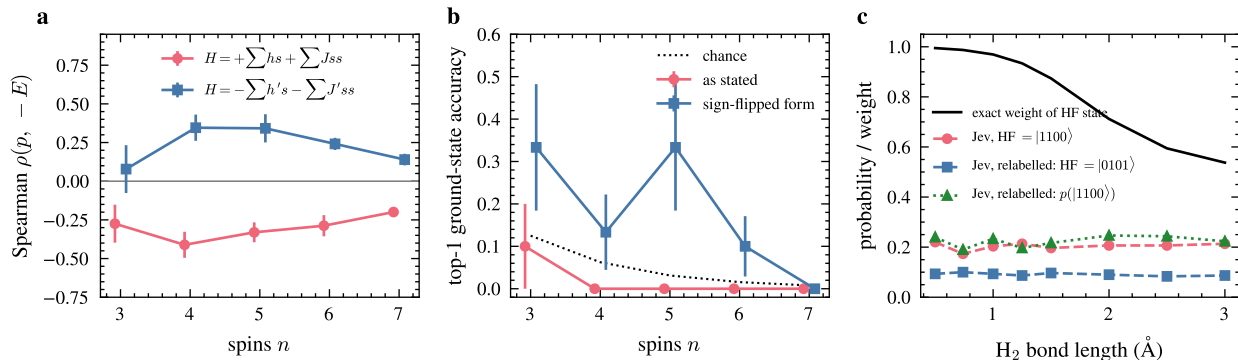


Figure 3: Jev given a Hamiltonian in its input (E2). **a**, Rank correlation between Jev’s probability for each spin configuration and its negative energy, for random Ising models as stated (circles) and with the same energies written in the sign-flipped textbook form (squares); mean $\pm$ s.e.m. over 10 instances. **b**, Top-1 ground-state accuracy; dotted line, chance 2^{-n} . **c**, Exact weight of the Hartree–Fock state of H₂ (STO-3G, Jordan–Wigner) and Jev’s probability for it, before and after relabelling the qubits so that the Hartree–Fock state is $|0101\rangle$.

with $h' = -h$ and $J' = -J$. The correlation became positive (+0.08 to +0.35), β became positive (0.03 to 0.22), and top-1 accuracy rose above chance for $n \leq 6$ (descriptively; ten instances per n). Every energy is unchanged, so Jev’s answer depends on how the Hamiltonian is printed.

The Jordan–Wigner Hamiltonian of H₂ in STO-3G separates reading from recall. Asked which of the 16 basis states carries the largest ground-state weight, Jev chose the Hartree–Fock state $|1100\rangle$ in 16 of 24 queries and $|0011\rangle$ otherwise, with a probability for $|1100\rangle$ near 0.2 at every bond length while the exact weight falls from 0.995 to 0.54 (Fig. 3c). It placed 17% of its probability on states outside the stated two-particle sector. When we relabelled the qubits so that the Hartree–Fock state becomes $|0101\rangle$, Jev still chose $|1100\rangle$ in 23 of 24 queries and gave the true dominant state a probability of 0.09: the answer comes from the textbook encoding, not from the printed Pauli terms. Its ground-state energy, read from a ten-level Score, had a mean absolute error of $104 mE_h$, worse than a constant guess at the median exact energy ($65 mE_h$); every one of the 24 read-outs had its mode at the highest level. The Score levels are $50 mE_h$ apart, so this instrument could not have shown chemical accuracy in any case; the informative comparison is with the constant.

4.3 E3: chemistry decisions track the stated stretch factor

Table 1 scores Jev’s marginal probabilities against the PySCF labels. Its rankings are informative (AUROC 0.76, 0.81 and 0.89), but every prompt states the stretch factor, and that number alone ranks the cases at least as well (0.92, 0.78 and 0.93); for multireference character Jev is worse than the bare stretch factor (difference -0.16 , 95% CI -0.28 to -0.02). Within the one stretch level where the labels are mixed (1.5 times equilibrium), Jev’s AUROC is 0.57, 0.58 and 0.48. In ablations added after review, removing the molecule name changed little (AUROC 0.83, 0.76, 0.90), whereas removing the stretch note while keeping name and geometry dropped Jev to 0.57, 0.65 and 0.71, with Brier scores worse than the base rate. Four rewordings of each question moved AUROC by at most 0.11 (the largest move raised multireference character to 0.87) and did not change this picture.

Jev’s probabilities are also poorly calibrated: its Brier score is much worse than that of a leave-one-molecule-out logistic model on the stretch factor and HOMO–LUMO gap for multireference character and instability, and equal for T_1 . Platt recalibration [58], fitted leave-one-molecule-out, closes the gap for instability and improves T_1 , but does not help for multireference character. Used

Table 1: Quantum-chemistry decisions on 48 cases (E3; T_1 on 47). All logistic and recalibration models are fitted leave-one-molecule-out. ^aAUROC of the unfitted stretch factor; Brier score of a leave-one-molecule-out logistic model on it. Molecule-clustered 95% intervals for AUROC are shown in Fig. 4a.

Predictor	Multireference		$T_1 > 0.02$		RHF unstable	
	AUROC	Brier	AUROC	Brier	AUROC	Brier
Base rate	–	0.236	–	0.234	–	0.182
Stretch factor (stated in prompt) ^a	0.92	0.123	0.78	0.189	0.93	0.110
Logistic (stretch, HOMO–LUMO gap)	0.89	0.118	0.79	0.171	0.90	0.104
Jev	0.76	0.207	0.81	0.170	0.89	0.184
Jev, Platt-recalibrated	0.73	0.199	0.80	0.151	0.86	0.112
Logistic + Jev	0.86	0.128	0.79	0.146	0.93	0.082
Jev, name removed	0.83	0.172	0.76	0.183	0.90	0.175
Jev, stretch note removed	0.57	0.329	0.65	0.306	0.71	0.338
Jev, geometry only	0.59	0.339	0.65	0.310	0.68	0.346

as an extra feature in the logistic model, Jev gives small but consistent Brier improvements for T_1 (-0.025 , 95% CI -0.038 to -0.011) and for instability (-0.023 , CI -0.042 to -0.002), and none for multireference character; given the stretch factor alone, adding Jev lowers the log loss significantly only for T_1 . So Jev carries a little information beyond two cheap descriptors, but most of what it reports is the stretch factor it was given.

4.4 E4: determinant ranking for selected CI is a fixed template

Selected CI keeps only the determinants that matter [15, 16]. For ten systems we built a symmetric active space on stable RHF orbitals (CAS(6e,5o) with 100 determinants for seven systems, 25 determinants for three), gave Jev the active orbital energies and the determinant list in occupation notation, and asked which non-reference determinant has the largest weight in the exact ground state. We added determinants in the resulting order, in spin-complete groups (all determinants with the same spatial occupation), diagonalised the Hamiltonian in each subspace, kept the lowest singlet root, and measured the error against full CASCI. Comparators were the oracle ordering by exact $|c_i|^2$, the first-order weight $|H_{i0}|^2/(H_{ii} - H_{00})^2$, the Epstein–Nesbet second-order contribution $|H_{i0}|^2/(H_{ii} - H_{00})$ used once, iterative CIPSI, which recomputes that criterion against the current wavefunction after each addition, excitation rank, and random order; all ties were broken at random and averaged over 30 draws.

Averaged over the whole curve, Jev’s ordering beats random order in 9 of 10 systems (paired Wilcoxon $p = 0.006$) and the excitation-rank rule in 9 of 10 ($p = 0.049$), is indistinguishable from the one-shot first- and second-order estimates (6 of 10 each, $p = 0.70$ and 0.85), and loses to iterative CIPSI in all ten ($p = 0.002$; Fig. 4b,c). By the metric that matches the question asked, Jev failed. Its top choice was never the oracle’s top determinant; in every system it was the HOMO→LUMO single excitation, which by Brillouin’s theorem does not couple to the RHF reference. Across the seven systems with the same active space, its ranking was nearly identical (median Spearman correlation 0.97 between systems), so Jev applies one template to every molecule. Asking instead one Noul per determinant (“weight above 0.01”) did worse and was indistinguishable from random order (7 of 10, $p = 0.43$).

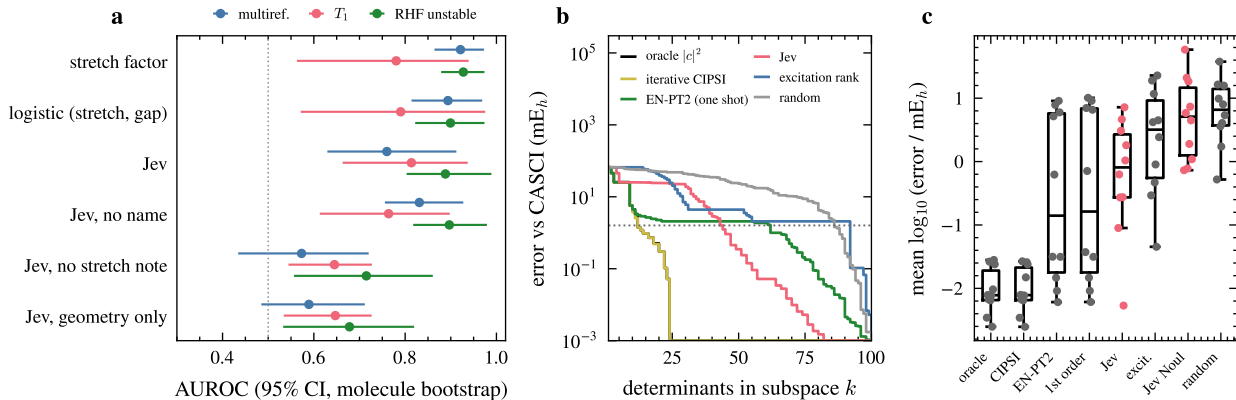


Figure 4: Jev in quantum-chemistry workflows. **a**, AUROC with 95% molecule-clustered bootstrap intervals for the three E3 labels, for the unfitted stretch factor, a logistic model, Jev, and three Jev input ablations. **b**, Selected-CI error against subspace size for N_2 at equilibrium, CAS(6e,5o)/cc-pVDZ; dotted line, $1.6 mE_h$. **c**, Mean of $\log_{10}(\text{error}/mE_h)$ over the whole curve, with errors floored at $10^{-3} mE_h$, for ten systems; lower is better. “Jev Noul” is the per-determinant elicitation.

4.5 E5–E6: a larger molecule set and a routing workflow (added after review)

To test whether E3 depends on textbook molecules, we repeated it on 31 less common molecules (among them LiF, NaH, MgO, SiS, P_2 , Be_2 , O_3 , FNO, HOF and N_2O), each at the same four stretches, with the same prompts and the same labelling protocol. Their equilibrium geometries were supplied from memory by the coding agent rather than taken from a database; labels and prompts use the same geometries, so the comparison is internally consistent, but “1.0 times equilibrium” is approximate. After excluding cases without a converged RHF, CCSD or singlet CASCI root, 111–114 of the 124 cases remained per label, and the results resemble E3 (Fig. 5a). Jev’s AUROC was 0.81, 0.76 and 0.80, against 0.87, 0.69 and 0.86 for the bare stretch factor and 0.84, 0.68 and 0.85 for the leave-one-molecule-out logistic model; none of the differences from the stretch factor is significant. With geometry only, AUROC fell to 0.64, 0.54 and 0.56. There is weak evidence that Jev draws on more than the stated stretch here: within single stretch levels its AUROC for multireference character is 0.76–0.90, but these strata contain as few as two cases of one class, and adding Jev to a model of the stretch factor lowers the log loss significantly only for instability. Its probabilities are also compressed, with the mean $p(\text{multireference})$ rising only from 0.42 to 0.58 across stretches while the true rate rises from 0.10 to 0.91. Adding Jev to the logistic model again reduced the Brier score slightly, significantly only for instability (-0.014 , 95% CI -0.024 to -0.002).

For 32 small species (atoms, carbenes, silylenes, and diatomics such as C_2 , SiC, BN and O_2 ; 16 singlets and 16 triplets by construction) we asked for the ground-state multiplicity, with CCSD(T)/cc-pVDZ adiabatic gaps as reference. Single-reference CCSD(T) is itself uncertain for C_2 , B_2 , SiC and BN, and for BN it disagrees with experiment. Jev was right for 26 of 32 (0.81), close to ordering the states by their Hartree–Fock energies (0.78) and above always answering singlet (0.50); five of its six errors called a singlet a triplet (Fig. 5b).

Finally we asked whether Jev can route calculations. Of 116 small cases where full CI is affordable, 19 were excluded, 16 of them because RHF, CCSD or full CI did not converge. In the remaining 97 (78 in STO-3G, 19 in 6-31G; all electrons correlated), CCSD(T) missed full CI by more than $1.6 mE_h$ in 38%. Jev’s probability that CCSD(T) is reliable ranked the failures with AUROC 0.76 (95% CI 0.65–0.85), against 0.65 for the T_1 diagnostic, which needs a CCSD calculation; the difference is not significant (CI -0.03 to 0.24). Counting the eight cases with

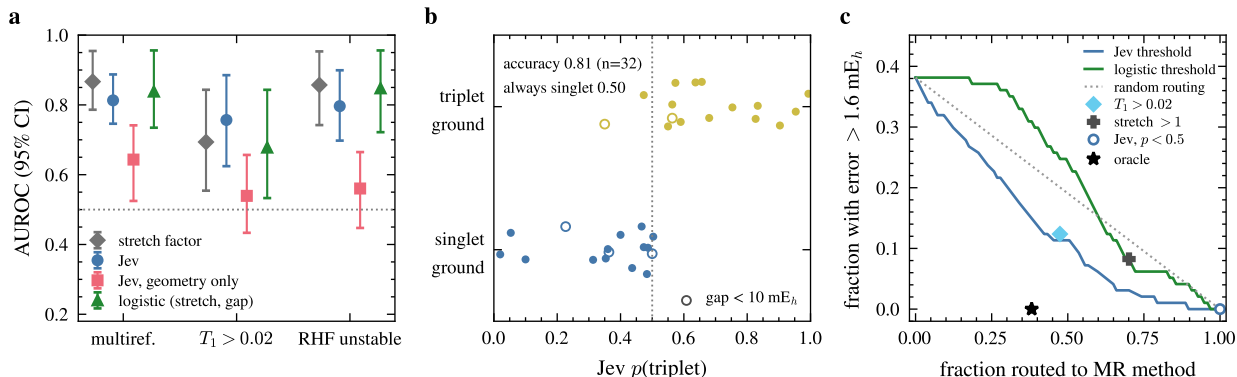


Figure 5: A larger set and a routing workflow (E5–E6). **a**, AUROC with molecule-clustered 95% intervals on 31 less common molecules for the unfitted stretch factor, Jev, Jev with geometry only, and a leave-one-molecule-out logistic model. **b**, Jev’s probability of a triplet ground state for 32 species, grouped by the CCSD(T) ground state; open markers have gaps below 10 mE_h . **c**, Fraction of cases whose final error exceeds 1.6 mE_h against the fraction routed to a multireference method, for thresholds on Jev and on the logistic model, the T_1 rule, a stretch rule, Jev’s own $p < 0.5$ threshold and an oracle.

unconverged CCSD as failures gives 0.78 and 0.64. A logistic model on stretch and HOMO–LUMO gap, fitted leave-one-molecule-out, was below chance (0.39). At the budget of the $T_1 > 0.02$ rule, which sends 47% of cases to a multireference method, Jev’s ranking left 11 failures against 12 for T_1 , and a smaller mean final error over all 97 cases (2.2 against 10.8 mE_h ; Fig. 5c). A rule that routes every stretched geometry leaves 8 failures but routes 70% of cases; at that budget Jev’s ranking leaves 3. These curves sweep thresholds on the same cases they are scored on, which is fair for the ranking but says nothing about where to put a threshold. Jev’s own probabilities cannot set the threshold: they lie between 0.15 and 0.46 while the true reliability rate is 0.62, so “route if $p < 0.5$ ” sends every case to the expensive method, and their Brier score (0.358) is worse than the base rate (0.236).

4.6 E7: open-weight language models (added after review)

To see whether these properties are specific to Jev, we repeated E1, the negation check, E3 and E2 on Qwen2.5-1.5B-Instruct and Qwen2.5-7B-Instruct [59], run locally, reading $p(\text{yes})$ from the next-token probabilities of “Yes” and “No” (which carried at least 99.98% of the probability mass on every prompt) and scoring Ising configurations by the log-probability of their labels. Both models were less coherent than Jev (Fig. 6a). Their median coupling asymmetry was 0.69 and 1.16 nats against 0.26 for Jev, the law of total probability failed by a median of 0.11 and 0.32 against 0.04, and a statement plus its negation summed to between 0.70 and 1.62 (1.5B) and between 0.00 and 1.69 (7B). The 1.5B model agreed with whatever fact it was told, giving positive couplings even to the pair “multireference” and “RHF adequate”, which chemically should be negative; the 7B model’s probabilities were so extreme that half of them fell outside our clipping range, so its asymmetry is sensitive to the clip (on the unclipped pairs its median is 0.39). On the E3 labels the 1.5B model was below chance and the 7B model was informative but badly calibrated (Fig. 6b). Neither open model read the Ising Hamiltonians at all: their rank correlations stayed within 0.1 of zero under both sign conventions and followed the position of options in the list (Fig. 6c). Jev is therefore more coherent than these general models, which is consistent with its training for calibrated decisions, and its sign-following is a response to the printed Hamiltonian that the open models do not show;

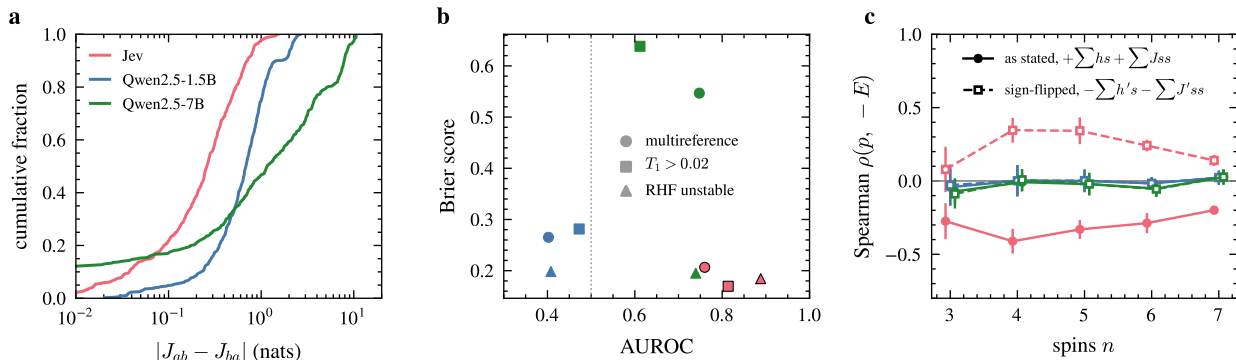


Figure 6: Jev against two open-weight models (E7). **a**, Cumulative distribution of coupling asymmetry over the 480 case-pair combinations of E1. **b**, AUROC and Brier score on the three E3 labels. **c**, Rank correlation between probability and negative energy for random Ising models written in both sign conventions; mean $\pm$ s.e.m. over instances.

neither kind of model reads the Hamiltonian as a physicist would.

5 Discussion

Jev’s outputs can be read as local energies, and within one domain stated facts shift them roughly additively, which lets single-fact answers predict two-fact answers. That additivity does not require a shared landscape, and the evidence against one is consistent: asymmetric couplings, violated total probability and unnormalised negations persist under a matched null, under rewording, and in a control where the exact joint is printed in the input. This is the behaviour one expects when heads are calibrated one at a time [21], since calibrated marginals need not share a joint. For incompatible conditionals of this kind, a Gibbs sampler converges to a distribution that depends on the update order [21], so there would be no single energy whose minima one could search for. Because Jev also adopts stated facts only partially, a cleaner separation of incoherence from soft conditioning would need access that the API does not provide.

Given a Hamiltonian as text, Jev does not compute with it. Its preferences follow the printed signs of the coefficients, reverse when the same energies are printed with opposite signs, and for H_2 return the textbook answer even after the qubits are relabelled. This agrees with the developer’s own warning that Jev is not a calculator [9] and with the fragility of language models on numerical reasoning [54]. A model whose ranking of the eight configurations of a three-spin Ising model reverses with the sign convention cannot stand in for an annealer or a variational solver [4, 32].

For quantum-chemistry decisions, most of Jev’s apparent skill came from a stretch factor that we stated, which is the information a chemist would also use, and it vanished when that note was removed. With the note, Jev adds a small, statistically clear amount of information to a two-descriptor model for two of three labels, at a cost of thousandths of a cent per query, which makes it a plausible extra feature in a triage model if it is recalibrated on local data. It should not be used to rank determinants: its ranking is one template shared across molecules, and iterative CIPSI, which a selected-CI code already runs, is far better. On less familiar molecules there was weak evidence of discrimination beyond the stretch note, and as a router Jev ranked CCSD(T) failures at least as well as T_1 without any correlated calculation, and better than a rule that routes every stretched geometry at the same budget. Both uses need a threshold chosen on local data, because Jev’s own probabilities are compressed toward the middle and shifted. The open-weight comparison

places Jev between two failures: it is far more coherent than general models read through their token probabilities, and it responds to a printed Hamiltonian where they do not, yet it responds to the wrong feature of it.

Limitations. We studied one model version through a black box, and later versions may behave differently. Our conditionals are soft, because stated facts are only partly adopted. The chemistry sets use uniform bond stretches that are stated in the prompt, E5 geometries are approximate, labels depend on thresholds and on the active space, the routing set is mostly in a minimal basis, the spin-state set is balanced by construction, and the selected-CI spaces are far from production scale. The open-weight comparison covers two small models of one family, read through token probabilities. Pairwise and three-way tests cover a small part of the possible coherence conditions, and the three-way test is underpowered. Two-decimal rounding and sampling noise limit the resolution of every energy we report.

6 Conclusion

Jev’s typed probabilities can be read as local energies that combine roughly additively, but they do not form one landscape, and they are not a Hamiltonian. They do not track a physical Hamiltonian supplied in the input, and in chemistry they mostly repeat what the prompt states. The black-box tests used here (coupling symmetry under a matched null, a stated-joint positive control, and held-out prediction by symmetric and asymmetric models) apply to any model that answers many questions about one input.

Code and data. Code, the pre-registered plan, all cached API responses and the PySCF ground truth are at <https://github.com/dongzhaohe321418-lab/jev-hamiltonian>. Every analysis and figure can be regenerated offline from the cache.

Use of AI tools. This study was carried out with extensive help from AI systems. Anthropic’s Claude, working as a coding agent, wrote the experiment and analysis code, ran the experiments, compiled the literature survey (each reference was checked against a primary source) and drafted the text. The pre-release review described in Section 3 was simulated by separate Claude agents and audited by OpenAI Codex; no human peer review has taken place. The author directed the study, checked the results against the released data and takes responsibility for the content.

References

- [1] Yann LeCun, Sumit Chopra, Raia Hadsell, Marc’Aurelio Ranzato, and Fu Jie Huang. A tutorial on energy-based learning. In Gökhan Bakır, Thomas Hofmann, Bernhard Schölkopf, Alexander J. Smola, and Ben Taskar, editors, *Predicting Structured Data*. MIT Press, 2006. URL <http://yann.lecun.com/exdb/publis/pdf/lecun-06.pdf>.
- [2] Will Grathwohl, Kuan-Chieh Wang, Jörn-Henrik Jacobsen, David Duvenaud, Mohammad Norouzi, and Kevin Swersky. Your classifier is secretly an energy based model and you should treat it like one, 2019. URL <https://arxiv.org/abs/1912.03263>.
- [3] David H. Ackley, Geoffrey E. Hinton, and Terrence J. Sejnowski. A learning algorithm for Boltzmann machines. *Cognitive Science*, 9(1):147–169, 1985. doi: 10.1207/s15516709cog0901_7.

- [4] Andrew Lucas. Ising formulations of many NP problems. *Frontiers in Physics*, 2:5, 2014. doi: 10.3389/fphy.2014.00005.
- [5] Tadashi Kadowaki and Hidetoshi Nishimori. Quantum annealing in the transverse Ising model. *Physical Review E*, 58(5):5355–5363, 1998. doi: 10.1103/PhysRevE.58.5355.
- [6] Diogo Almeida. Introducing system one models & Jev. TypeSafe AI Blog, September 2026. URL <https://typesafe.ai/blog/introducing-system-one-models-and-jev>. Published 2026-09-15.
- [7] TypeSafe AI. API reference. TypeSafe AI documentation, 2026. URL <https://docs.typesafe.ai/api>.
- [8] Simon Willison. Jev introduces a new shape of LLM—system one, aka decision models. Blog post, simonwillison.net, September 2026. URL <https://simonwillison.net/2026/Sep/21/jev/>. Published 2026-09-21.
- [9] TypeSafe AI. Jev 1.13 jaggedness. TypeSafe AI documentation, 2026. URL <https://docs.typesafe.ai/model-jaggedness/jev-1.13>.
- [10] Julian Besag. Spatial interaction and the statistical analysis of lattice systems. *Journal of the Royal Statistical Society: Series B (Methodological)*, 36(2):192–225, 1974. doi: 10.1111/j.2517-6161.1974.tb00999.x.
- [11] Barry C. Arnold and S. James Press. Compatible conditional distributions. *Journal of the American Statistical Association*, 84(405):152–156, 1989. doi: 10.1080/01621459.1989.10478750.
- [12] P. Jordan and E. Wigner. Über das Paulische äquivalenzverbot. *Zeitschrift für Physik*, 47(9–10):631–651, 1928. doi: 10.1007/BF01331938.
- [13] P. J. J. O’Malley, R. Babbush, I. D. Kivlichan, J. Romero, J. R. McClean, R. Barends, J. Kelly, P. Roushan, A. Tranter, N. Ding, B. Campbell, Y. Chen, Z. Chen, B. Chiaro, A. Dunsworth, A. G. Fowler, E. Jeffrey, E. Lucero, A. Megrant, J. Y. Mutus, M. Neeley, C. Neill, C. Quintana, D. Sank, A. Vainsencher, J. Wenner, T. C. White, P. V. Coveney, P. J. Love, H. Neven, A. Aspuru-Guzik, and J. M. Martinis. Scalable quantum simulation of molecular energies. *Physical Review X*, 6(3):031007, 2016. doi: 10.1103/PhysRevX.6.031007.
- [14] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.
- [15] B. Huron, J. P. Malrieu, and P. Rancurel. Iterative perturbation calculations of ground and excited state energies from multiconfigurational zeroth-order wavefunctions. *The Journal of Chemical Physics*, 58(12):5745–5759, 1973. doi: 10.1063/1.1679199.
- [16] Adam A. Holmes, Norm M. Tubman, and C. J. Umrigar. Heat-bath configuration interaction: An efficient selected configuration interaction algorithm inspired by heat-bath sampling. *Journal of Chemical Theory and Computation*, 12(8):3674–3680, 2016. doi: 10.1021/acs.jctc.6b00407.
- [17] J. J. Hopfield. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8):2554–2558, 1982. doi: 10.1073/pnas.79.8.2554.

- [18] Hubert Ramsauer, Bernhard Schäfl, Johannes Lehner, Philipp Seidl, Michael Widrich, Thomas Adler, Lukas Gruber, Markus Holzleitner, Milena Pavlović, Geir Kjetil Sandve, Victor Greiff, David Kreil, Michael Kopp, Günter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need, 2020. URL <https://arxiv.org/abs/2008.02217>.
- [19] Mike Winer and Boris Hanin. Deep neural nets as Hamiltonians, 2025. URL <https://arxiv.org/abs/2503.23982>.
- [20] J. M. Hammersley and P. Clifford. Markov fields on finite graphs and lattices. Unpublished manuscript, 1971. URL <https://ora.ox.ac.uk/objects/uuid:4ea849da-1511-4578-bb88-6a8d02f457a6>.
- [21] David Heckerman, David Maxwell Chickering, Christopher Meek, Robert Rounthwaite, and Carl Kadie. Dependency networks for inference, collaborative filtering, and data visualization. *Journal of Machine Learning Research*, 1:49–75, 2000. URL <https://www.jmlr.org/papers/v1/heckerman00a.html>.
- [22] Lukas Fluri, Daniel Paleka, and Florian Tramèr. Evaluating superhuman models with consistency checks, 2023. URL <https://arxiv.org/abs/2306.09983>.
- [23] Daniel Paleka, Abhimanyu Pallavi Sudhir, Alejandro Alvarez, Vineeth Bhat, Adam Shen, Evan Wang, and Florian Tramèr. Consistency checks for language model forecasters. In *International Conference on Learning Representations (ICLR)*, 2025. URL <https://openreview.net/forum?id=r5IXBlTCGc>.
- [24] Jian-Qiao Zhu and Thomas L. Griffiths. Incoherent probability judgments in large language models, 2024. URL <https://arxiv.org/abs/2401.16646>. CogSci 2024.
- [25] Nora Kassner, Oyvind Tafjord, Hinrich Schütze, and Peter Clark. BeliefBank: Adding memory to a pre-trained language model for a systematic notion of belief. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 8849–8861, 2021. doi: 10.18653/v1/2021.emnlp-main.697.
- [26] Diego Calanzone, Stefano Teso, and Antonio Vergari. Towards logically consistent language models via probabilistic reasoning, 2024. URL <https://arxiv.org/abs/2404.12843>. ICLR 2024 Workshop on Reliable and Responsible Foundation Models.
- [27] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, 2017. URL <https://proceedings.mlr.press/v70/guo17a.html>.
- [28] Mahdi Pakdaman Naeini, Gregory Cooper, and Milos Hauskrecht. Obtaining well calibrated probabilities using Bayesian binning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 29, 2015. doi: 10.1609/aaai.v29i1.9602.
- [29] Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950. doi: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.
- [30] Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam

- Bowman, Stanislav Fort, Deep Ganguli, Danny Hernandez, Josh Jacobson, Jackson Kernion, Shauna Kravec, Liane Lovitt, Kamal Ndousse, Catherine Olsson, Sam Ringer, Dario Amodei, Tom Brown, Jack Clark, Nicholas Joseph, Ben Mann, Sam McCandlish, Chris Olah, and Jared Kaplan. Language models (mostly) know what they know, 2022. URL <https://arxiv.org/abs/2207.05221>.
- [31] Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, Rafael Rafailov, Huaxiu Yao, Chelsea Finn, and Christopher D. Manning. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 5433–5442, 2023. doi: 10.18653/v1/2023.emnlp-main.330.
 - [32] Alberto Peruzzo, Jarrod McClean, Peter Shadbolt, Man-Hong Yung, Xiao-Qi Zhou, Peter J. Love, Alán Aspuru-Guzik, and Jeremy L. O’Brien. A variational eigenvalue solver on a photonic quantum processor. *Nature Communications*, 5:4213, 2014. doi: 10.1038/ncomms5213.
 - [33] Rongxin Xia, Teng Bian, and Sabre Kais. Electronic structure calculations and the Ising Hamiltonian. *The Journal of Physical Chemistry B*, 122(13):3384–3395, 2018. doi: 10.1021/acs.jpcc.7b10371.
 - [34] Scott N. Genin, Ilya G. Ryabinkin, and Artur F. Izmaylov. Quantum chemistry on quantum annealers, 2019. URL <https://arxiv.org/abs/1901.04715>.
 - [35] Giuseppe Carleo and Matthias Troyer. Solving the quantum many-body problem with artificial neural networks. *Science*, 355(6325):602–606, 2017. doi: 10.1126/science.aag2302.
 - [36] David Pfau, James S. Spencer, Alexander G. D. G. Matthews, and W. M. C. Foulkes. Ab initio solution of the many-electron Schrödinger equation with deep neural networks. *Physical Review Research*, 2(3):033429, 2020. doi: 10.1103/PhysRevResearch.2.033429.
 - [37] Jan Hermann, Zeno Schätzle, and Frank Noé. Deep-neural-network solution of the electronic Schrödinger equation. *Nature Chemistry*, 12(10):891–897, 2020. doi: 10.1038/s41557-020-0544-y.
 - [38] Sam McArdle, Suguru Endo, Alán Aspuru-Guzik, Simon C. Benjamin, and Xiao Yuan. Quantum computational chemistry. *Reviews of Modern Physics*, 92(1):015003, 2020. doi: 10.1103/RevModPhys.92.015003.
 - [39] Yudong Cao, Jonathan Romero, Jonathan P. Olson, Matthias Degroote, Peter D. Johnson, Mária Kieferová, Ian D. Kivlichan, Tim Menke, Borja Peropadre, Nicolas P. D. Sawaya, Sukin Sim, Libor Veis, and Alán Aspuru-Guzik. Quantum chemistry in the age of quantum computing. *Chemical Reviews*, 119(19):10856–10915, 2019. doi: 10.1021/acs.chemrev.8b00803.
 - [40] H. Chau Nguyen, Riccardo Zecchina, and Johannes Berg. Inverse statistical problems: from the inverse Ising problem to data science. *Advances in Physics*, 66(3):197–261, 2017. doi: 10.1080/00018732.2017.1341604.
 - [41] Nathan Wiebe, Christopher Granade, Christopher Ferrie, and D. G. Cory. Hamiltonian learning and certification using quantum resources. *Physical Review Letters*, 112(19):190501, 2014. doi: 10.1103/PhysRevLett.112.190501.
 - [42] Anurag Anshu, Srinivasan Arunachalam, Tomotaka Kuwahara, and Mehdi Soleimanifar. Sample-efficient learning of interacting quantum systems. *Nature Physics*, 17:931–935, 2021. doi: 10.1038/s41567-021-01232-0.

- [43] Timothy J. Lee and Peter R. Taylor. A diagnostic for determining the quality of single-reference electron correlation methods. *International Journal of Quantum Chemistry*, 36(S23):199–207, 1989. doi: 10.1002/qua.560360824.
- [44] Wanyi Jiang, Nathan J. DeYonker, and Angela K. Wilson. Multireference character for 3d transition-metal-containing molecules. *Journal of Chemical Theory and Computation*, 8(2): 460–468, 2012. doi: 10.1021/ct2006852.
- [45] Elvira R. Sayfutyarova, Qiming Sun, Garnet Kin-Lic Chan, and Gerald Knizia. Automated construction of molecular active spaces from atomic valence orbitals. *Journal of Chemical Theory and Computation*, 13(9):4063–4078, 2017. doi: 10.1021/acs.jctc.7b00128.
- [46] WooSeok Jeong, Samuel J. Stoneburner, Daniel King, Ruye Li, Andrew Walker, Roland Lindh, and Laura Gagliardi. Automation of active space selection for multireference methods via machine learning on chemical bond dissociation. *Journal of Chemical Theory and Computation*, 16(4):2389–2399, 2020. doi: 10.1021/acs.jctc.9b01297.
- [47] J. P. Coe. Machine learning configuration interaction. *Journal of Chemical Theory and Computation*, 14(11):5739–5749, 2018. doi: 10.1021/acs.jctc.8b00849.
- [48] Chenru Duan, Fang Liu, Aditya Nandy, and Heather J. Kulik. Data-driven approaches can overcome the cost–accuracy trade-off in multireference diagnostics. *Journal of Chemical Theory and Computation*, 16(7):4373–4387, 2020. doi: 10.1021/acs.jctc.0c00358.
- [49] Fang Liu, Chenru Duan, and Heather J. Kulik. Rapid detection of strong correlation with machine learning for transition-metal complex high-throughput screening. *The Journal of Physical Chemistry Letters*, 11(19):8067–8076, 2020. doi: 10.1021/acs.jpcllett.0c02288.
- [50] Jon Paul Janet and Heather J. Kulik. Predicting electronic structure properties of transition metal complexes with neural networks. *Chemical Science*, 8(7):5137–5152, 2017. doi: 10.1039/C7SC01247K.
- [51] Andres M. Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D. White, and Philippe Schwaller. Augmenting large language models with chemistry tools. *Nature Machine Intelligence*, 6(5):525–535, 2024. doi: 10.1038/s42256-024-00832-8.
- [52] Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624(7992):570–578, 2023. doi: 10.1038/s41586-023-06792-0.
- [53] Adrian Mirza, Nawaf Alampara, Sreekanth Kunchapu, Martiño Ríos-García, Benedict Emoekabu, Aswanth Krishnan, Tanya Gupta, Mara Schilling-Wilhelmi, Macjonathan Okereke, Anagha Aneesh, et al. A framework for evaluating the chemical knowledge and reasoning abilities of large language models against the expertise of chemists. *Nature Chemistry*, 17(7):1027–1034, 2025. doi: 10.1038/s41557-025-01815-x.
- [54] Iman Mirzadeh, Keivan Alizadeh, Hooman Shahrokhi, Oncel Tuzel, Samy Bengio, and Mehrdad Farajtabar. GSM-Symbolic: Understanding the limitations of mathematical reasoning in large language models, 2024. URL <https://arxiv.org/abs/2410.05229>.
- [55] Qiming Sun, Xing Zhang, Samragni Banerjee, Peng Bao, Marc Barbry, Nick S. Blunt, Nikolay A. Bogdanov, George H. Booth, Jia Chen, Zhi-Hao Cui, et al. Recent developments in the PySCF

- program package. *The Journal of Chemical Physics*, 153(2):024109, 2020. doi: 10.1063/5.0006074.
- [56] Jarrod R. McClean, Nicholas C. Rubin, Kevin J. Sung, Ian D. Kivlichan, Xavier Bonet-Monroig, Yudong Cao, et al. OpenFermion: the electronic structure package for quantum computers. *Quantum Science and Technology*, 5(3):034014, 2020. doi: 10.1088/2058-9565/ab8ebc.
- [57] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.
- [58] Hsuan-Tien Lin, Chih-Jen Lin, and Ruby C. Weng. A note on Platt’s probabilistic outputs for support vector machines. *Machine Learning*, 68(3):267–276, 2007. doi: 10.1007/s10994-007-5018-6.
- [59] Qwen Team. Qwen2.5 technical report, 2024. URL <https://arxiv.org/abs/2412.15115>.